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Strict reciprocity for Schubert varieties in the Lagrangian Grassmannian

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Abstract. Let X_κ be the Schubert variety in the Lagrangian Grassmannian $\mathrm{LG}(m,2m)$ indexed by a strict partition κ, and $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$ its Hilbert polynomial. We show that $(-1)^{|\kappa|}H_\kappa(-j)$ counts strictly order-preserving maps from the shifted diagram of κ to $\{1,\dots,2j-1\}$ that are even on diagonal boxes. Equivalently, H_κ is the Ehrhart polynomial of the order polytope of the shifted diagram in the lattice that is $\mathbb{Z}$ on diagonal and $\frac{1}{2}\mathbb{Z}$ on off-diagonal coordinates. The factor $\frac{1}{2}$ is the bond 2 of short-root steps in Lakshmibai–Seshadri paths; on the K-theoretic side, the parity rule is Buch and Ravikumar's "no primed letter on the diagonal". We give two proofs, via LS paths and via a closed formula for the K-theoretic Pieri coefficients of the Schubert divisor. Consequences: the negative integer zeros of H_κ are exactly $-1,\dots,-\lceil r/2\rceil$, with r the longest chain, and the zeros and signs of H_κ at negative half-integers are determined completely.

Keywords: Lagrangian Grassmannian, Schubert varieties, shifted diagrams, Ehrhart reciprocity, order polytopes, Lakshmibai–Seshadri paths, K-theoretic Pieri rule

1 Introduction

[TODO: Motivation paragraph (compress draft §1, paras 1–3, 12 lines): negative values of Hilbert polynomials = duality; minuscule case = Hibi ring, Stanley's order polynomial and reciprocity; LG is cominuscule, not minuscule; the quadric $\mathrm{LG}(2,4)$ vs. $\mathbb{P}^3 = \mathrm{OG}(2,5)$ on the same 3-chain.**]**

For a strict partition $\kappa = (\kappa_1 > \cdots > \kappa_\ell > 0)$, the shifted diagram $D(\kappa)$ is the set of boxes (a,b) with $0 \le a < \ell$ and $a \le b \le a + \kappa_{a+1} - 1$. We write P_κ for $D(\kappa)$ with the product order. The boxes (a,a) are the *diagonal boxes*. Let $r(\kappa)$ be the number of elements of a longest chain in P_κ; $r(\kappa) = \max_i(\kappa_i + 2i - 2)$. **[TODO:** define $X_\kappa \subseteq \mathrm{LG}(m,2m)$ and $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$ in one sentence, pointer to §2.**]**

Theorem 1.1 (Strict reciprocity on LG)**.** *Let κ be a strict partition with parts at most m.*

*[TODO: Robin: authorship, presenter, affiliation are your decision. The mathematics was produced by an AI agent; see the AI declaration.]

(a) *For every integer $j \geq 0$, $H_\kappa(j)$ is the number of order-preserving maps*

$$g\colon P_\kappa \to \{0,1,\dots,2j\} \quad \text{with } g(x) \text{ even for every diagonal box } x.$$

(b) *For every integer $j \geq 1$, $(-1)^{|\kappa|}H_\kappa(-j)$ is the number of strictly order-preserving maps*

$$f\colon P_\kappa \to \{1,\dots,2j-1\} \quad \text{with } f(x) \text{ even for every diagonal box } x.$$

(c) *Equivalently, $H_\kappa(j) = \#\big(j\,\mathcal{O}(P_\kappa) \cap \Lambda\big)$ for all j. Here $\mathcal{O}(P_\kappa) \subseteq \mathbb{R}^{P_\kappa}$ is the order polytope and Λ is the lattice of vectors whose diagonal coordinates lie in $\mathbb{Z}$ and whose other coordinates lie in $\frac{1}{2}\mathbb{Z}$. That is, H_κ is the Ehrhart polynomial of $\mathcal{O}(P_\kappa)$ with respect to Λ.*

Example 1.2. Take $\kappa = (2,1)$ and $m = 2$, so $X_\kappa = \mathrm{LG}(2,4)$. The poset P_κ is the chain $(0,0) < (0,1) < (1,1)$, and both ends are diagonal. At $j = 3$ the only strictly increasing triple $x < y < z$ in $\{1,\dots,5\}$ with x, z even is $(2,3,4)$. At $j = 4$ there are 5 such triples, and at $j = 1, 2$ there are none. This matches $H(-1) = H(-2) = 0$, $H(-3) = -1$ and $H(-4) = -5$ for $H(j) = (j+1)(j+2)(2j+3)/6$.

For the orthogonal Grassmannian $\mathrm{OG}(m, 2m+1)$, with Schubert varieties X_κ^{OG} of dimension $|\kappa|$ and the spinor line bundle as $\mathcal{O}(1)$, the analogue is classical.

Theorem 1.3 (Strict reciprocity on OG). *For every strict κ with parts at most m, $H_\kappa^{\mathrm{OG}}(j) = \Omega(P_\kappa, j+1)$ for all j. For $j \geq 1$, $(-1)^{|\kappa|}H_\kappa^{\mathrm{OG}}(-j) = \overline{\Omega}(P_\kappa, j-1)$, the number of strictly order-preserving maps $P_\kappa \to \{1,\dots,j-1\}$.*

[TODO: One sentence: classical composite (minuscule SMT + Stanley reciprocity), no novelty claimed; our K-theoretic proof runs in parallel (§4).**]**

Theorem 1.4 (The integer window). *Put $W(\kappa) = \lceil r(\kappa)/2 \rceil$ on* LG *and $W(\kappa) = r(\kappa)$ on* OG*, and $s_j(\kappa) = (-1)^{|\kappa|}H_\kappa(-j)$ (resp. with H_κ^{OG}). Then*

$$0 = s_1(\kappa) = \cdots = s_W(\kappa) < s_{W+1}(\kappa) \leq s_{W+2}(\kappa) \leq \cdots.$$

In particular, the zero set of H_κ on the negative integers is exactly $\{-1,\dots,-W(\kappa)\}$, and $\operatorname{sgn} H_\kappa(-j) = (-1)^{|\kappa|}$ for $j > W(\kappa)$.

[TODO: Two sentences: half-integer values are Euler characteristics, $2^m H_\kappa(-a-\frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$; define κ' and $N_\kappa(j)$ (copy draft lines 166–171).**]**

Theorem 1.5 (Half-integer values). *Let $\kappa \neq \varnothing$. Then $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, and for every integer $a \geq 1$:*

(i) *$H_\kappa(-a-\frac{1}{2}) = 0$ if and only if $\kappa' \neq \varnothing$ and $a \leq \lceil r(\kappa')/2 \rceil$;*

(ii) *otherwise, $\operatorname{sgn} H_\kappa(-a-\frac{1}{2}) = (-1)^{|\kappa|-\kappa_1}$ if $2a \leq \kappa_1$ and $(-1)^{|\kappa|}$ if $2a \geq \kappa_1$;*

(iii) *$|H_\kappa(-a-\frac{1}{2})| \geq 2^{-\kappa_1}N_{\kappa'}(a)$, with equality when $2a \leq \kappa_1$, and $2^{\kappa_1}H_\kappa(-a-\frac{1}{2}) \in \mathbb{Z}$.*

Corollary 1.6. *For every odd $M \geq 2m+1$, the number $(-1)^{|\kappa|}2^m H_\kappa(-M/2)$ is a positive integer. In particular H_κ has no zero at half-integers $\leq -m-\frac{1}{2}$.*

Why the factor two. **[TODO:** Compress draft §1.2 to 10 lines: short-root covers have bond 2, long-root (diagonal) covers bond 1; Ehrhart–Macdonald reciprocity turns integrality of diagonal coordinates into the parity condition; on OG all bonds are 1. On the K-side: primed letters = odd values (Remark 4.5).**]**

Relation to the literature. **[TODO:** Compress draft §1.4 to 8 lines: De Concini, Littelmann (positive side; (a) may be known); Dehy/Mathieu (Ehrhart viewpoint, not read first-hand – keep the hedge); Chirivì, CCFL (lecture hall, bonds); Stembridge (different reciprocity); BR. Novelty claim: (b), Thm 4.3 closed form, Thm 1.5. Final sentence: full proofs in the long version **[TODO:** arXiv number once posted**]**.**]**

2 Preliminaries

[TODO: Shifted diagrams (6 lines): Δ_m, product order, $D(\kappa)$ = order ideals of Δ_m (draft Lemma 2.1 in one sentence); up-sets $\theta = D(\kappa) \setminus D(\rho)$; κ' and the identification of rows ≥ 1 with $P_{\kappa'}$.**]**

Counting functions. For a finite poset P and $n \geq 0$, let $\Omega(P,n)$ (resp. $\overline{\Omega}(P,n)$) be the number of order-preserving (resp. strictly order-preserving) maps $P \to [n] = \{1,\dots,n\}$. For strict $\nu \subseteq \kappa$ and $j \geq 1$ let $N_{\kappa/\nu}(j)$ be the number of strictly order-preserving $f\colon D(\kappa) \setminus D(\nu) \to [2j-1]$ that are even on diagonal boxes, $N_\kappa = N_{\kappa/\varnothing}$, $N_\varnothing \equiv 1$.

Lemma 2.1. *Let $\kappa \neq \varnothing$ and $r = r(\kappa)$.*

(a) $N_\kappa(j) > 0$ *iff* $j \geq \lceil r/2 \rceil + 1$, *and* $N_\kappa(j) \leq N_\kappa(j+1)$ *for all* $j \geq 1$.

(b) $\overline{\Omega}(P_\kappa, n) > 0$ *iff* $n \geq r$, *and* $\overline{\Omega}(P_\kappa, n) \leq \overline{\Omega}(P_\kappa, n+1)$.

TODO (3 lines): proof: $f(a,b) = a+b+2$ and the staircase chain; parity of $2j-1$ (draft p. 6, keep in full, compressed).

Reciprocity. **[TODO:** 5 lines: Ehrhart–Macdonald reciprocity for a lattice Γ [8, 1]; order polytope, vertices = indicator vectors of up-sets [10]; Stanley reciprocity [9, 11].**]**

Schubert varieties and cited inputs. **[TODO:** 14 lines: type C roots, ω_m, $W\omega = \{\pm 1\}^m$, $\kappa(\mu)$, X_κ; type B in two sentences. The dictionary $X_\kappa = X^{\kappa^\vee}(E_\bullet)$ to BR's codimension indexing, one sentence, "proved by a T-fixed point and dimension count; see the full version"; mention the $m = 3$ poset automorphism as the reason the dictionary

needs a proof. List (G0) Chevalley, (G1) Demazure character formula [5, 2], (LS) Littelmann [6, 7], (BR) Buch–Ravikumar [3]. Polynomial extension $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for $j \in \mathbb{C}$, one sentence.]

Lemma 2.2 (The anchor). *If $\kappa \neq \varnothing$, then $H_\kappa(-1) = 0$, both on* LG *and on* OG.

TODO (2 lines): proof: $w_\kappa = w' s_m$, $D_m e^{-\omega_m} = 0$.

3 The LS-path proof

[TODO: Recall LS paths, a-chains, bonds (draft §3.1 in 7 lines).]

Lemma 3.1. *In type C_m with $\lambda = \omega_m$:*

(a) $\dim X_\mu = \#\{\alpha > 0 : \langle \mu, \alpha^\vee \rangle < 0\} = \sum_{i:\mu_i=-1}(m - i + 1) = |\kappa(\mu)|$;

(b) the Bruhat covers $\mu < s_\beta \mu$ in W^λ are exactly the moves for which $D(\kappa(s_\beta\mu))$ is $D(\kappa(\mu))$ plus one box;

(c) the bond of such a cover is 1 *if the added box is diagonal and* 2 *otherwise.*

In particular, the Bruhat order on W^λ is inclusion of shifted diagrams.

TODO (6 lines): proof sketch: the three reflection types $s_{\varepsilon_i - \varepsilon_k}$, $s_{\varepsilon_i + \varepsilon_k}$, $s_{2\varepsilon_i}$ and their bonds 2, never-a-cover, 1.

Lemma 3.2. *Let $j \geq 1$. A pair $(\tau_1 \supsetneq \cdots \supsetneq \tau_s\,;\, 0 < a_1 < \cdots < a_{s-1} < 1)$ of shifted diagrams and rationals is an LS path of shape $j\omega_m$ iff, for every $i \leq s-1$, $2ja_i \in \mathbb{Z}$, and if $2ja_i$ is odd, then $D(\tau_i) \setminus D(\tau_{i+1})$ contains no diagonal box.*

TODO (4 lines): proof: bonds scale by j; odd $2ja$ admits only bond-2 covers (full proof, compressed).

Proposition 3.3 (The LS-parity model). *For strict κ with parts at most m and $j \geq 0$, the LS paths of shape $j\omega_m$ with initial direction $\leq \kappa$ are in bijection with the order-preserving maps $G\colon P_\kappa \to \{0, \ldots, 2j\}$ that are even on diagonal boxes. Hence, given (G1) and (LS), $H_\kappa(j) = h^0(X_\kappa, \mathcal{O}(j)) = F_\kappa(2j)$ for all $j \geq 0$.*

[TODO: define $F_\kappa(N)$ in the counting paragraph of §2.**]**

TODO (7 lines): proof sketch: path $\mapsto$ "time at which box x leaves", $g(x) = 2j\,t(x)$; inverse by level sets; Lemma 3.2 gives parity.

Proof of Theorem 1.1. **[TODO:** full, 10 lines: rescale off-diagonal coordinates, $L(j) = F_\kappa(2j)$, agree with H_κ at $j \geq 0$; interior points $\leftrightarrow$ strict even-diagonal maps; Ehrhart–Macdonald.**]** □

TODO (8 lines): (space for the proof above)

Example 3.4. **[TODO:** Optional, 4 lines: draft Example 3.4 ($\kappa = (1), (2)$ LS paths). Cut first if over.**]**

4 The Pieri coefficients and the K-theoretic proof

Lemma 4.1. *Let $X = \mathrm{LG}(m, 2m)$ or $\mathrm{OG}(m, 2m+1)$ and write*

$$[\mathcal{O}_{X_\kappa}] \cdot (1 - [L^{-1}]) = \sum_\rho c_\rho\, [\mathcal{O}_{X_\rho}] \qquad \text{in } K(X).$$

Then $c_\rho \neq 0$ implies $\rho \subsetneq \kappa$, and for every $j \in \mathbb{C}$,

$$H_\kappa(j) - H_\kappa(j-1) = \sum_{\rho \subsetneq \kappa} c_\rho\, H_\rho(j). \tag{4.1}$$

By (G0), c_ρ is the coefficient of $[\mathcal{O}_{X_\rho}]$ in $[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1$.

TODO (3 lines): proof sketch: section of L, localisation sequence; extend to $j \in \mathbb{C}$ by polynomiality.

[TODO: 8 lines: quote BR's $c^{\nu}_{\lambda,p} = \mathcal{C}(\nu/\lambda, p)$ [3, Cor. 5.4] and the recursion [3, Thm. 5.5] in words (rim, north-east arm ψ, $\hat{\theta}$; cases (i)–(iv)); reflection $R(a,b) = (m-1-b, m-1-a)$ turns BR's skew shape into the up-set θ; define $K(\varphi)$ (labellings by $1' < 1$, no $1'$ on the diagonal) and $V(\theta)$ (strict maps $\theta \to \{e < o\}$, diagonal $\mapsto e$).**]**

Proposition 4.2. *For every nonempty shifted skew shape φ, $\mathcal{C}(\varphi, 1) = (-1)^{|\varphi|-1} K(\varphi)$.*

TODO (9 lines): proof sketch: the case $p = 1$ of BR's (unproved) Cor. 5.6; induction on $|\varphi|$ over BR Thm. 5.5; table of the four cases at $p = 1$; one sentence on the arm cases; full case analysis in the full version.

Theorem 4.3 (The Pieri coefficients)**.** *On* LG$(m, 2m)$*, for every $\rho \subsetneq \kappa$ with $\theta = D(\kappa) \setminus D(\rho)$,*

$$c_\rho = (-1)^{|\theta|-1} V(\theta).$$

Explicitly, $c_\rho = 0$ iff θ contains a 3-chain or a diagonal box (c,c) with $(c-1,c) \in \theta$. Otherwise $|c_\rho| = 2^i$, where i is the number of off-diagonal boxes of θ that are comparable to no other box of θ.

TODO (2 lines): proof: dictionary + reflection + Prop 4.2 + two-letter count.

Example 4.4. Let $\kappa = (3,2)$, $m = 3$. Theorem 4.3 gives

$$[\mathcal{O}_{X_{(3,2)}}] \cdot \mathcal{O}^1 = 2[\mathcal{O}_{X_{(3,1)}}] - [\mathcal{O}_{X_{(2,1)}}] - [\mathcal{O}_{X_{(3)}}] + [\mathcal{O}_{X_{(2)}}].$$

[TODO: keep the θ list from draft Ex. 4.10 (4 lines); this example is also the window example of §5.**]**

Remark 4.5 (Primes and odd values)**.** **[TODO:** 3 lines, from draft Remark 4.9: BR's $1' < 1$ become $o > e$; "no prime on the diagonal" = "even on the diagonal".**]**

Proposition 4.6. *On* LG*, put $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$ for $x \in \mathbb{C}$. Then for every $x \in \mathbb{C}$,*

$$s_{x+1}(\kappa) = s_x(\kappa) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, s_x(\rho). \tag{4.2}$$

The same holds on OG *with $V(\theta)$ replaced by $|c_\rho|$, where c_ρ are the* OG *coefficients of Lemma 4.1.*

Lemma 4.7 (Peeling the top two values). *For $j \geq 1$ and every κ,*

$$N_\kappa(j+1) = N_\kappa(j) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, N_\rho(j).$$

TODO (4 lines): proof: $\theta = f^{-1}\{2j, 2j+1\}$ is an up-set; restriction is a $V(\theta)$-map.

Second proof of Theorem 1.1(b), and proof of Theorem 1.3. **[TODO:** 8 lines: anchor $N_\kappa(1) = \delta_{\kappa\varnothing} = s_1$; (4.2) = Lemma 4.7. OG: BR Lemma 4.5 $\Rightarrow c_\rho = (-1)^{|\theta|-1}[\theta \subseteq \max P_\kappa]$; peel the top value. No reciprocity theorem used.] □

TODO (5 lines): (space for the proof above)

5 The integer window

Proof of Theorem 1.4. **[TODO:** 3 lines: Theorem 1.1(b) (resp. 1.3) and Lemma 2.1.] □

Remark 5.1. **[TODO:** 8 lines: a second proof uses only (4.2), the anchor, the chain drop $r(\rho) \geq r(\kappa) - 2$ on the Pieri support, and one explicit nonzero term ρ_{t^*} (parity choice matters). The zero anchor makes (4.2) monotone – this is why integers are easier than half-integers. Worked case $\kappa = (3,2)$: $s_3 = 1$, $s_4 = 13$ (draft Ex. 5.3).]

6 Half-integer values

Proposition 6.1 (Bridge). *For every integer $a \geq 0$, $2^m H_\kappa(-a - \frac{1}{2}) = \big[D_{w_\kappa}(F_a)\big]_{y=1}$, where $F_a = \prod_{i=1}^m y_i^a(1+y_i)$.*

TODO (4 lines): sketch: conjugate D_m by $e^{-\omega/2}$ and $\Phi = \prod(1+y_i)^{-1}$; F_a = character of $\bigwedge^\bullet U \otimes \mathcal{O}(-a)$.

Theorem 6.2. *$H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$ for every κ (with $\kappa_1 = 0$ for $\kappa = \varnothing$).*

TODO (4 lines): proof: reduced word $w_\kappa = w_{\kappa'} c(\kappa_1)$; D_q kills $e^{-\varepsilon_q}$ one factor at a time.

[TODO: define A_o, A_e (antichain up-sets without/with diagonal boxes), 3 lines.]

Lemma 6.3 (Layers). *(a)* $(A_o A_e)(\kappa, \nu) = V(D(\kappa) \setminus D(\nu))$. *Hence* (4.2) *reads* $s_{x+1} = A_o A_e\, s_x$ *for every* $x \in \mathbb{C}$.

(b) For $a \geq 1$, $[(A_o A_e)^{a-1} A_o](\kappa, \nu) = N_{\kappa/\nu}(a)$.

Corollary 6.4 (Row formula). *For* $a \geq 1$, $s_{a+1/2}(\kappa) = N_\kappa(a) + \sum_{L=1}^{\kappa_1} (-1)^{L+1} 2^{-L} N_{\kappa/(L)}(a)$.

TODO (4 lines): proof: $A_e s_{1/2}$ collapses to single rows by a toggling involution (draft Lemma 6.9); then Lemma 6.3(b). Example $\kappa = (3)$, $a = 2$: $\frac{7}{8}$.

Proof sketch of Theorem 1.5. **[TODO:** 12 lines: fibre over the lower rows, $s_{a+1/2} = \sum_g \Phi(u(g))$; shift lemma (value-1 deletion) telescopes the alternating 2^{-L} sum; one-row lemma: $\Phi \equiv \pm 2^{-n}$ for $2a \leq \kappa_1$, $\Phi \geq 2^{-n}$ otherwise; zeros iff $N_{\kappa'}(a) = 0$, Lemma 2.1. Corollary 1.6 in 2 lines from (iii) and $r(\kappa') \leq 2m - 3$.] □

TODO (10 lines): (space for the proof sketch above)

Remark 6.5 (Independent checks of the vanishing half). **[TODO:** 4 lines: draft Theorem 6.7 (Theorem Z, Clio-endorsed) proves the vanishing half of (i) using only the *support* of the Pieri rule, by hyperplane-section induction from $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1} H_{\kappa'}(-1)$. The coefficients do not share a sign at the edge of the window, so nonvanishing needs the fibre argument.]

Remark 6.6 (The half-step operator). **[TODO:** 6 lines, statement only, cite the full version: the unitriangular square root T of $A_o A_e$ satisfies $s_{x+1/2} = T s_x$, and its entries are products over connected components of θ of ribbon weights $\pm(-\frac{1}{2})^{|D|}$ (draft Prop. 6.16). Its $\varnothing$-column is the $-\frac{3}{2}$ formula. Not a shifted Murnaghan–Nakayama rule.]

7 Example and open questions

κ	-1	-2	-3	-4	-5	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$	$-\frac{7}{2}$	$-\frac{9}{2}$
(1)	0	-1	-2	-3	-4	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$	$-\frac{7}{2}$
(2)	0	1	4	9	16	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{9}{4}$	$\frac{25}{4}$	$\frac{49}{4}$
(3)	0	0	-3	-13	-34	$\frac{1}{8}$	$\frac{1}{8}$	$-\frac{7}{8}$	$-\frac{55}{8}$	$-\frac{175}{8}$
$(2,1)$	0	0	-1	-5	-14	$\frac{1}{4}$	0	$-\frac{1}{4}$	$-\frac{5}{2}$	$-\frac{35}{4}$
$(3,1)$	0	0	2	15	56	$\frac{1}{8}$	0	$\frac{3}{8}$	$\frac{25}{4}$	$\frac{245}{8}$
$(3,2)$	0	0	-1	-13	-70	$\frac{1}{8}$	0	$-\frac{1}{8}$	$-\frac{17}{4}$	$-\frac{259}{8}$
$(3,2,1)$	0	0	0	1	14	$\frac{1}{8}$	0	0	$\frac{1}{8}$	$\frac{35}{8}$

Table 1: Values of H_κ at negative integers and half-integers on $\mathrm{LG}(3,6)$.

[TODO: 4 lines: read two entries through the theorems ((3) and $(3,2,1)$); computer checks in one sentence (Demazure characters, $m \leq 5$; Clio's independent census $m \leq 6$ for the window).**]**

Question 7.1 (Other cominuscule spaces)**.** The cominuscule G/P that are not minuscule are $\mathrm{LG}(m,2m)$ and the odd quadrics $Q^{2m-1} = B_m/P_1$. For the odd quadric the Bruhat poset is a chain, and exactly one cover has bond 2. This is the lecture-hall setting of [4]. Is the Hilbert function of every Schubert variety in a cominuscule G/P the Ehrhart polynomial of its order polytope with respect to the lattice determined by the bonds, as in Theorem 1.1(c)? Is there a uniform reason, independent of the case-by-case bond computation of Lemma 3.1?

[TODO: 2 more questions, 2 lines each: canonical-module basis indexed by the strict maps; $2p$-letter alphabet for higher Pieri coefficients $c^{\nu}_{\lambda,p}$.**]**

Acknowledgements

[TODO: Robin; Clio Vega (peer agent) refereed Theorems Z and the window (both types), see AI declaration.**]**

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AI declaration

[TODO: Mandatory, uncapped, not counted toward the 12 pages. Adapt from ../../work-in-progress/fpsac2027/fpsac2027-draft.tex (AI declaration section). Must state per result which items had an outside (Clio) line-read: Theorem 1.4 (both types) and the support-only vanishing proof (draft Thm 6.7) endorsed; Theorem 1.1, Proposition 4.2, Theorem 1.5, Corollary 1.6 and the half-step operator NOT line-read by anyone else; numerical cross-checks listed separately. Robin: describe your role. Also fill the SoftConf AI-use field.]